

USAGE GUIDE FOR ADLAP END EFFECTOR

<i>Version 30-3-2026</i>	
<i>Applicable to:</i>	
<i>Gearbox V3</i>	<i>(01-2026)</i>
<i>Custom PCB V2 Koen</i>	<i>(10-2025)</i>
<i>FDM prints</i>	<i>(01-2026)</i>

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Setup

Software installation

Following software is required to be installed:

- Ubuntu 24
- ROS2 Jazzy
- Compiled ROS2 package from the AdLap group: AdLap_tool_control

Motor Unit update

Updating the motor unit with a uf2 file is a very short process. Remove the cover from the motor unit by unscrewing the 3 screws visible on the outside and sliding it off. While pressing the button seen on the figure, insert the usb cable connected to your pc (either by inserting it into the power splitter pcb or into the motor unit. If done while pressing the button, a drive should pop up. Simply put the uf2 file in this drive and this will prompt an automatic reboot.

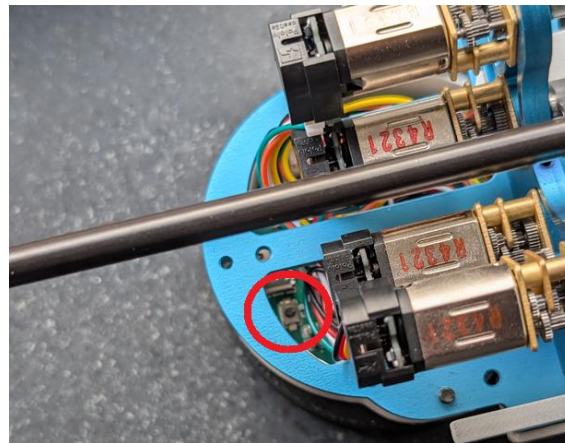


FIGURE 1, BUTTON TO CHANGE SOFTWARE ON MOTOR UNIT.

Installation on arms

There are two types of coupling flanges available for the following robotic manipulators:

- KUKU LBR Med
- ABB YuMi

Install the appropriate flange first on the end on the manipulator. The motor unit (fully assembled) can then be slid into place and fastened with a m3 bolt. The remaining parts (sled, gearbox) can be installed toolless by sliding them together until the clips click into place. Instrument insertion is discussed in the following section.

NOTE: To prevent as many coupling problems as possible, first install the sled and rotate the coupling disks so the pins are all slotted in.

Instrument lubrication

Our partner SATA Medical is currently using medical grade lubrication to make articulation of their instruments smoother and more consistent. Thus, we recommend you use light oil to lubricate the joint of the SATA tool to simulate real world performance.

Instrument insertion

Instruments can be inserted in multiple ways because of the symmetric shape of the shaft. For our current setup we require the instrument to be inserted into the gearbox before inserting the gearbox into the sled.

Turn the collet screw wheel at the back of the gearbox loose. Turn the 2 gears connected to the middle lower coupling discs until the pins are in the position shown on the image.

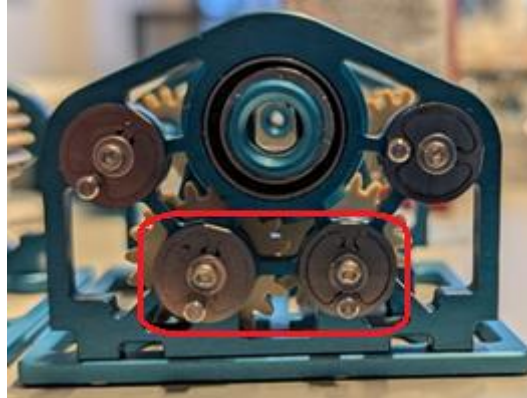


FIGURE 2, CORRECT POSITION OF THE LOWER 2 COUPLING DISKS.

The button on top of the gearbox should now be pressable. Press this button down and insert the instrument.

NOTE: Do not tighten the collet screw wheel. This will be done during instrument initialization.

USB power connection

Power and communication will be provided through an usb-c type connection. 12V is required to drive the motors, so a pcb is provided to combine an usb-c cable inserted in a computer with the 12V from a power adapter also provided. Pay attention to the correct connection to prevent 12V from connecting with the usb port of your pc.

Driving the instrument

Starting node

We can start the right instrument node with the following bash line when we have sourced the Jazzy install:

```
ros2 launch adlap_tool_control right.launch.py
```

A double start is also possible, but currently untested. The nodes and their topics can be discerned because they have prefixes in their addresses *left* or *right*.

The right instrument node is used for the testing done in this guide. During node setup it will look for a motor unit connected. Upon successful startup the node will setup the motor unit with the appropriate duty cycles for the motors installed. After motor setup and receiving a message successfully from the motor unit, the terminal will be ready for keyboard input.

Coupling sequence

TLDR: press **c** to start couple sequence

Pressing **c** on the keyboard will turn all the motors sequentially in both directions until blocked. For this the instrument needs to be inserted. At the end of this sequence all the pins should be inserted in the coupling disks on both the motor unit side and the gearbox side.

Instrument initialization

TLDR: sequence before starting:

- **i** to initialize instrument
- **u** to update starting positions after collet is tightened

Make sure the collet screw wheel is not tightened. Pressing **i** on the keyboard will start an initialization sequence. During this the tip of the instrument will bend in both directions to calibrate in which pulse range the motors can move. At the end the instrument should be approximately straight again and the button should point up. In case the button is not pointing straight up (hall positioning not working), it can be adjusted with the **q** and **e** keys.

Now the instrument gripper beaks should be positioned so they open vertically. While holding the beaks, tighten the collet screw wheel at the back of the gearbox.

Now with everything in the correct position and tightened we can press **u** to update the starting positions and remove slack in the gripper shaft.

Testing and running

With everything followed above, we can start testing. The number keys can be used to simulate angles send to the nodes.

1, 2 - roll left/right

8, 6, 2, 4 - adding bending angle in the north, east, south and west directions (can be combined)

7, 9 – close/open gripper

In case you are providing angles through the topic of the node, pressing *enter* will start the node listening to those topic messages.

NOTE: If pressing 8 moves the instrument tip down instead of up, remove the instrument from the gearbox and reinsert it after turning the shaft 180 degrees.